Corporate Finance

Topic 3

Bernardo Ricca

Hidden information

- In many markets, on side has better information than the other
- Examples:
  - Sellers of used cars know more about the quality of their cars than potential buyers
  - Clients seeking for insurance know more than the insurance providers about their accident risk
  - Workers know more than employers about their abilities
  - Borrowers know more than lenders about their default probability, the value of assets in place and pledged collateral, and the private benefit associated with different investment choices
  - In an IPO some (connected) investors know more about the firm than others

Some potential implications

- **Adverse selection**: at a given interest rate, only riskier borrowers are willing to apply for a loan
- **Cross-subsidization**: low-risk borrowers have to pay higher interest rates due to the presence of high-risk borrowers
  - High-risk borrowers might obtain financing and invest in negative *NPV* projects due to the subsidy they receive (**overinvestment**)
- Extreme case: **market breakdown**
  - Credit rationing can arise: even positive *NPV* projects ate not financed (**underinvestment)**

A model of financing under asymmetric information

- A penniless borrower (entrepreneur, firm) needs I to invest in a project that generates R in case of success and 0 in case of failure
- (A1) The borrower and the outside investors are **risk-neutral** $\Rightarrow$ all future payoffs are discounted at the risk-free rate (risk premium = 0)
- (A2) The risk-free interest rate is normalized to zero: $r_f = 0$
- (A1) + (A2) $\Rightarrow$ discount rate for any future payoff is $r = 0$

$$r = \text{risk premium} + r_f = 0$$

- Capital markets are competitive and outside investors have access to a risk-free bond that yields a zero net return

Private information about default probabilities

- The entrepreneur can be one of two types (privately known):
  - Good type (low default risk): probability of success p
  - Bad type (high default risk): probability of success q

  where $1 > p > q > 0$

- The entrepreneur has private information about his type: outside investors cannot tell bad and good borrowers apart
- Outside investors assign probability $\alpha \in (0, 1)$ on the borrower being a good type
  - We can interpret α as the share of good types in a large population

Efficiency

- We assume that $I \leq pR$: if the borrower is a **good type**, the project has a **positive** *NPV*

- From a social surplus perspective, the good type should be able to obtain financing and undertake the project: he/she can transform the investment of I into a higher value

$$\underbrace{I}_{\text{investment}} \leq PV(\textit{expected future cash flows}) = \frac{pR}{1+r} = pR$$

Efficiency

- Regarding the **bad type**, we will study two cases
- **(i)** $I \leq qR$: the project also has a **positive** *NPV*
  - From a social surplus perspective, the bad type should be able to undertake the project
- **(ii)** $I > qR$: the project has a **negative** *NPV*
  - From a social surplus perspective, the bad type should **not** be able to undertake the project, as it destroys value

$$\underbrace{I}_{\text{investment}} > PV(\textit{expected future cash flows}) = \frac{qR}{1+r} = qR$$

Efficiency

- When $qR < I$, it would be more efficient if the bank denies credit to the borrower and invests at the zero risk-free interest rate (a zero *NPV* transaction, instead of a negative one):

- Investing I at the zero risk-free rate:

$$PV(\textit{expected future cash flows}) = \frac{I}{1+r} = I$$

- Investing I in the project:

$$PV(\textit{expected future cash flows}) = \frac{qR}{1+r} = qR < I$$

Private information about default probabilities

- Under asymmetric information, outside investors assign a probability of success of

$$m = \alpha p + (1 - \alpha) q$$

- As m is an average of p and q and $\alpha \in (0, 1)$, $p > m > q$
- When $qR \geq I$, $mR \geq I$ (for any α)
- When $qR < I$, we might have $mR \geq I$ or $mR < I$, depending on α:

$$mR \geq I \Leftrightarrow \alpha \geq \frac{I/R - q}{p - q} \quad \text{and} \quad mR < I \Leftrightarrow \alpha < \frac{I/R - q}{p - q}$$

Contracts

- The borrower is protected by limited liability, that is, the borrower cannot pay the lender out of his/her own wealth
- Outside (uninformed) investors receive $R_u \leq R$ if the project succeeds and zero if the project fails
- The entrepreneur (firm) receives $R_f = R - R_u$ in case of success and zero otherwise

Benchmark: symmetric information

- Symmetric information: the lender can observe the entrepreneur's type (default/repayment probability)
- Two different credit markets (one for each type of entrepreneur), with two different equilibrium contracts
  - A market for good type entrepreneurs, with an equilibrium contract establishing a repayment R_u^G to outside investors in case of success if the transaction takes place
  - A market for bad type entrepreneurs, with an equilibrium contract establishing a repayment R_u^B to outside investors in case of success if the transaction takes place

Good type contract: symmetric information

- The good type entrepreneur obtains financing and the project has a positive NPV
- As capital markets are competitive, the payment to outside investors in case of success satisfies (break-even condition)

$$pR_u^G = I \iff R_u^G = \frac{I}{p}$$

- The expected payoff (utility) of the entrepreneur is $pR_f^G = p(R - R_u^G) = pR - I$, that is, the entrepreneur receives the entire NPV

Bad type contract: symmetric information

- If $qR < I$, the bad type entrepreneur does not obtain financing as the outsider investors cannot break even:
  - The maximum expected amount lenders can obtain (known as the expected pledgeable income), qR, is smaller than I
- If $qR \geq I$, the bad type entrepreneur obtains financing and competitive capital markets imply that

$$qR_u^B = I \iff R_u^B = \frac{I}{q}$$

Moreover, the expected payoff (utility) of the entrepreneur is $qR_f^B = q(R - R_u^B) = qR - I$, that is, the entrepreneur receives the entire NPV

Benchmark: symmetric information

- Note that, as $p > q$, the good type repays a smaller amount in case of success

$$\underbrace{R_u^G}_{\text{repayment good type}} = \frac{I}{p} < \frac{I}{q} = \underbrace{R_u^B}_{\text{repayment low type}}$$

- Alternatively, if we interpret the contract as a debt contract, the good type pays a lower interest rate

$$i_G = \frac{R_u^G - I}{I} = \frac{1}{p} - 1 < \frac{1}{q} - 1 = \frac{R_u^B - I}{I} = i_B$$

- The outcome under symmetric information is **efficient** (maximizes social surplus) as only value-creating, $NPV > 0$, projects are financed

Asymmetric information

- We now assume again that the repayment probability is private information
- Is symmetric-information outcome, with one contract for each type, implementable?
- No. The bad type entrepreneur would lie and pretend to be a good entrepreneur in order to repay R_u^G instead of R_u^B
- As all types choose the repayment R_u^G, the outside investors' expected profit would be negative

$$\underbrace{m}_{\alpha p+(1-\alpha)q} R_u^G - I \;<\; pR_u^G - I \;=\; 0$$

Asymmetric information

- As lenders cannot distinguish between good and bad borrowers, there is only one equilibrium contract for both types that requires a repayment of R_u in case of success
- As $R_u \leq R$ because of limited liability, $R_f = R - R_u \geq 0$ and both the good and the bad types prefer this contract and investing in the project to not being financed (even if the project has a negative NPV)
- Outside investors' expected profit is

$$mR_u - I = [\alpha p + (1 - \alpha)q]R_u - I$$

- In particular, the maximum profit of outside investors is $mR - I$

Market breakdown and underinvestment

- If $mR < I$, lenders cannot break even and there is no lending: **market breakdown**
- This case arises whenever $qR < I$ and the probability that the borrower is a good borrower is low enough, that is, whenever $\alpha < \alpha^*$, where

$$[\alpha^* p + (1 - \alpha^*)q]R = I \quad \Rightarrow \quad \alpha^* = \frac{I/R - q}{p - q}$$

- **Underinvestment**: the good borrower does not invest even though the project has a positive *NPV*. The outcome is **inefficient**
- The good borrower is hurt because of the presence of bad types and the fact that outside investors suspect that the good type might be a bad type

Cross-subsidization

- If $mR \geq I$, there is lending
- Competition implies zero expected profits, that is,

$$mR_u = I \iff R_u = \frac{I}{m}$$

- As $q < m < p$, $R_u^G < R_u < R_u^B$ (**cross-subsidization**): the good type pays more in comparison to the symmetric information case ("fair value"), there is dilution and they receive less than the NPV
- Bad-type borrowers, on the other hand, pay less in comparison to the symmetric information case
- Alternatively, outside investors make money on the good type and lose money on the bad type
- Good type is again negatively affected by the presence of bad types, although to a lesser extent than in a market breakdown

Overinvestment

- When $mR \geq I$ and $qR \geq I$, there is cross-subsidization but the outcome is still **efficient**: only positive *NPV* projects are undertaken
- When $mR \geq I$ but $qR < I$, there is **overinvestment**: bad borrower invests even though his/her project has a negative *NPV* (the outcome is **inefficient**)
- There is adverse selection in the sense that the quality of entrepreneurs who obtain financing is worse than in the symmetric information benchmark

Cost incurred by the good type

- In the asymmetric-information case, when $mR > I$, the borrower's expected payoff is

$$pR_f = p\left(R - \frac{I}{m}\right) = \underbrace{(pR - I)}_{\substack{\text{symmetric information} \\ \text{payoff}}} - \underbrace{\frac{\chi}{1-\chi} I}_{\substack{\text{discount due to} \\ \text{information asymmetry}}}$$

where

$$\chi = \underbrace{(1-\alpha)}_{\text{probability of bad type}} \left(\frac{p-q}{p}\right)$$

- χ is a measure of adverse selection: the higher its value, the higher the discount due to information asymmetry

Example

- $R = 50$, $I = 30$, $p = 0.8$, $q = 0.7$, $\alpha = 0.5$
- The project has a positive NPV regardless of the borrower's type:
  - Good type: $NPV = -30 + 0.8 \times 50 = 10$
  - Bad type: $NPV = -30 + 0.7 \times 10 = 5$

Example

- **Symmetric information**: both types invest as projects have positive NPV
- Fair prices (no cross-subsidization), expected utility equal to the NPV:
  - Good type: $R_u^G = 30/0.8 = 37.5$, $i_G = 25\%$, $U_G = 0.8(50 - 37.5) = 10$
  - Bad type: $R_u^B = 30/0.7 = 42.9$, $i_B = 42.9\%$, $U_B = 0.7(50 - 42.9) = 5$
  - Expected discounted value of future cash flows:
    $\alpha p\ 50 + (1 - \alpha)q50 = 37.5$

Example

- **Asymmetric information**: lenders assign probability of success $m = 0.5 \times 0.8 + 0.5 \times 0.7 = 0.75$
- Both types invest, as $mR \geq I$
- Both types promise to repay $R_u = 40$, $i = 33.3\%$
- **Cross-subsidization**:
  - Good type has expected utility $U_G = 0.8(50 - 40) = 8$ (**smaller** than in the symmetric information case)
  - Bad type has expected utility $U_B = 0.7(50 - 40) = 7$ (**larger** than in the symmetric information case)
- Expected discounted value of future cash flows: $\alpha p\ 50 + (1 - \alpha)q50 = 37.5$ (**same** as in the symmetric information case $\Rightarrow$ outcome is still **efficient**)

Example

- Now, let's assume that $q = 0.5$
- The project has a negative *NPV* if the borrower is a bad type: $NPV = -30 + 0.5 \times 50 = -5$
- **Symmetric information**: only good type invests, same repayments and interest rates as before; bad type does not obtain credit (expected utility zero)
- Expected discounted value of future cash flows:
  - Since lenders do not fund the bad type, they invest 30 at the zero risk-free rate, obtaining 30 in the next period with probability 1
  - The expected discounted value is then: $\alpha p\ 50 + (1 - \alpha)30 = 35$

Example

- **Asymmetric information**: lenders assign probability of success $m = 0.5 \times 0.8 + 0.5 \times 0.5 = 0.65$
- Both types invest, as $mR \geq I$
- Both types promise to repay $R_u = 46.15$, $i = 53.8\%$
- **Cross-subsidization**:
  - Good type has expected utility $U_G = 0.8(50 - 46.15) = 3.1$ (**smaller** than in the symmetric information case, 10)
  - Bad type has expected utility $U_B = 0.5(50 - 46.15) = 1.9$ (**larger** than in the symmetric information case, 0)
- Expected discounted value of future cash flows: $\alpha p\ 50 + (1 - \alpha)q50 = 32.5$ (**smaller** than in the symmetric information case $\Rightarrow$ outcome is **inefficient (overinvestment)**)

Example

- What if $\alpha = 0.2$ and $q = 0.5$?
- **Symmetric information**: only good type invests, same repayments and interest rates as before; bad type does not obtain credit (expected utility zero)
  - The expected discounted value of future cash flows is then: $\alpha p\ 50 + (1 - \alpha)30 = 32$
- Asymmetric information: no credit (**market breakdown**), as $m = 0.2 \times 0.8 + 0.8 \times 0.6 = 0.56$ and $mR < I$
  - Both types obtain zero expected utility
  - Expected value of discounted cash flows is 30 (all money in the risk-free bond, smaller than in the symmetric information case $\Rightarrow$ outcome is **inefficient (underinvestment)**)

Example

- Recall that α^* is such that $[\alpha^* 0.8 + (1 - \alpha^*)0.5]R = I$, which yields $\alpha^* = 1/3$
- In general, we have two cases under asymmetric information:
  - Case 1: if the proportion of good types satisfies $\alpha \geq 1/3$, $mR \geq I$, both types invest, cross-subsidization and overinvestment
  - Case 2: if the proportion of good types satisfies $\alpha < 1/3$, $mR < I$, neither type invests, market breakdown and underinvestment

More information: the introduction of a credit bureau

- Suppose lenders can now observe a characteristic x, where either $x = x_G$ (good information) or $x = x_B$ (bad information)
  - Example: credit bureau that mandates the reporting of defaults and delinquencies; x_B would mean a checkered credit history, while x_G would mean a clean credit history
- Let's assume that $R = 50$, $I = 30$, $p = 0.8$, $q = 0.7$
- We assume that $P(x_G) = P(x_B) = \alpha = 0.5$
- In absence of the public mandatory signal x, the lenders' belief about the probability of success is:

$$m = \alpha p + (1 - \alpha)q = 0.5 \times 0.8 + 0.5 \times 0.7 = 0.75$$

More information: the introduction of a credit bureau

- Good types ($p = 0.8$) are more likely to have good information x_G:
  - The probability of observing good information x_G given that the borrower is a good type is $P(x_G|good\ type) = 0.9$
- Bad types ($q = 0.7$) are more likely to have bad information x_B:
  - The probability of observing bad information x_B given that the borrower is a bad type is $P(x_B|bad\ type) = 0.9$
- As probabilities add up to one, $P(x_B|good\ type) = 0.1$ and $P(x_G|bad\ type) = 0.1$

More information: the introduction of a credit bureau

- After observing x_G lenders update their belief about a borrower being a good type using the Bayes' rule

$$P(\textit{good type}|x_G) = \frac{P(x_G|\textit{good type}) \times P(\textit{good type})}{P(x_G)} = \frac{0.9 \times \alpha}{0.5} = 0.9$$

- Therefore, $P(\textit{bad type}|x_G) = 0.1$ and the belief about the probability of success conditional on observing x_G is now

$$\begin{aligned} m_{x_G} &= P(\textit{good type}|x_G) \times p + P(\textit{bad type}|x_G) \times q \\ &= 0.9 \times 0.8 + 0.1 \times 0.7 = 0.79 > 0.75 = m \end{aligned}$$

More information: the introduction of a credit bureau

- Similarly, after observing x_B lenders update their belief about a borrower being a good type using the Bayes' rule

$$P(\textit{good type}|x_B) = \frac{P(x_B|\textit{good type}) \times P(\textit{good type})}{P(x_B)} = \frac{0.1 \times \alpha}{0.5} = 0.1$$

- Therefore, $P(\textit{bad type}|x_B) = 0.9$ and the belief about the probability of success conditional on observing x_B is now

$$\begin{aligned} m_{x_B} &= P(\textit{good type}|x_B) \times p + P(\textit{bad type}|x_B) \times q \\ &= 0.1 \times 0.8 + 0.9 \times 0.7 = 0.71 < 0.75 = m \end{aligned}$$

More information: the introduction of a credit score

- Now we have "two" markets: one for x_G and one for x_B:
- One equilibrium repayment R_{x_G} conditional on observing x_G such that:
  - If $Rm_{x^G} \geq I$, both types that have information x_G receive credit and $R_{x_G} m_{x_G} = I$
  - Otherwise, if $Rm_{x_G} < I$, the market collapses (neither type receives credit)
- In the example, $m_{x_G} R = 0.79 \times 50 = 39.5 > 30$ and thus $R_{x_G} = 30/0.79 = 37.97$ and $i_{x_G} = 26.58\%$

More information: the introduction of a credit score

- One equilibrium repayment R_{x_B} conditional on observing x_B such that:
  - If $Rm_{x^B} \geq I$, both types that have information x_B receive credit and $R_{x_B} m_{x_B} = I$
  - Otherwise, if $Rm_{x_B} < I$, the market collapses (neither type receives credit)
- In the example, $m_{x_B} R = 0.71 \times 50 = 35.5 > 30$ and thus $R_{x_G} = 30/0.71 = 42.25$ and $i_{x_B} = 40.85\%$

More information: the introduction of a credit score

Information structure	Receives credit?	Interest rate	
		Good type	Bad type
Symmetric	Both types	25.0%	42.9%
Asymmetric (no bureau)	Both types	33.3%	33.3%
Asymmetric (good inf. x_G)	Both types	26.6%	26.6%
Asymmetric (bad inf. x_B)	Both types	40.8%	40.8%

More information: the introduction of a credit score

- The expected utility of the good type is

$$P(x_G|\textit{good type}) \times 9.62 + P(x_B|\textit{good type}) \times 6.2 = 9.27 > 8$$

- The expected utility of the bad type is

$$P(x_G|\textit{bad type}) \times 8.42 + P(x_B|\textit{bad type}) \times 5.42 = 5.72 < 7$$

- Less cross-subsidization: good type is better off in comparison to the situation without a credit bureau

- Ex ante expected value of future cash flows:

$$P(x_G) \times m_{x_G} R + P(x_B) \times m_{x_B} R = 37.5$$

same as before (no efficiency gains)

More information: the introduction of a credit score

- Can more information also improve efficiency?
- Consider the case again that $R = 50$, $I = 30$, $p = 0.8$, $q = 0.5$

Information structure	Receives credit?	Interest rate	
		Good type	Bad type
Symmetric	Both types	25.0%	No credit
Asymmetric (no bureau)	Both types	53.8%	53.8%
Asymmetric (good inf. x_G)	Both types	29.9%	29.9%
Asymmetric (bad inf. x_B)	Both types	No credit	No credit

More information: the introduction of a credit score

- Ex ante expected value of future cash flows:
  - Symmetric information: $0.5 \times p50 + 0.5 \times 30 = 35$
  - Asymmetric information (no bureau): $0.5 \times p50 + 0.5 \times q50 = 32.5$
  - Asymmetric information (with bureau): $P(x_G) \times m_{x_G} R + P(x_B) \times 30 = 34.5 > 32.5$
- The introduction of a credit bureau improves the capital allocation in the economy

Pecking-order hypothesis

- To minimize adverse selection costs, firms prefer to use internal funds to finance the projects
- If, in the model above, the entrepreneurs had wealth $A < I$, the good type would choose to borrow only $I - A$ to minimize the adverse selection discount
- To avoid being detected, the bad type would follow suit
- What about external funding? The pecking-order hypothesis posits that debt is preferred to equity
- The reason is that debt is less information-sensitive than equity

Pecking-order hypothesis

- An intuitive way to minimize cross-subsidization is to issue risk-free or over-collateralized debt

- Holmstrom (2015):

  Trading in debt that is sufficiently over-collateralized is a cheap way to avoid adverse selection. When both parties know that there is enough collateral, more precise private information about the collateral becomes irrelevant and will not impair liquidity. We saw this in the pawn contract. It did not matter that the pawnbroker's valuation of the watch was different from the borrower's. It sufficed that the pawnbroker felt confident that he could recover the loan by selling the watch, while the borrower was protected by the right to redeem the watch.

Pecking-order hypothesis

- To be able to distinguish debt and equity contracts let's suppose that the project pays R^S in case of success and R^F in case of failure
- Assumptions are the same as before: no assets in place (internal funding is not possible), the good type entrepreneur has a probability of success of p, the bad type entrepreneur has a probability of success of q, the investment costs I, the market believes that the share of good types is α, and the prior about the probability of success is $m = \alpha p + (1 - \alpha)q$
- We assume that $pR^S + (1 - p)R^F > qR^S + (1 - q)R^F > I$ (both projects have a positive NPV)

Pecking-order hypothesis

- The payoff of the projects can be split into a riskless and a risky part

$$pR^S + (1-p)R^F = \underbrace{R^F}_{\text{riskless}} + p\underbrace{(R^S - R^F)}_{\text{risky}}$$

$$qR^S + (1-q)R^F = \underbrace{R^F}_{\text{riskless}} + q\underbrace{(R^S - R^F)}_{\text{risky}}$$

- There is no asymmetric information about the riskless part

Pecking-order hypothesis

- If $R^F > I$, both parties can issue riskless debt: asymmetric information about the probability of success is irrelevant
- Let's assume that $R^F < I$
- We will see that the contract that minimizes cross-subsidization (and thus is the preferred choice of good types) involves pledging all the riskless (or information-insensitive) part to investors

Pecking-order hypothesis

- Contracts divide cash flows subject to limited liability:
  - In case of success, the entrepreneur receives R_f^S and investors receive $R_u^S = R^S - R_f^S$
  - In case of failure, the entrepreneur receives R_f^F and investors receive $R_u^F = R^F - R_f^F$

Pecking-order hypothesis

- To reap the benefits of cross-subsidization, bad-type entrepreneurs will want to pool (choose to issue the same contract) as the good-type
- Therefore, any contract $\{R_u^S, R_u^F\}$ must satisfy the investors' break-even condition:

$$mR_u^S + (1 - m)R_u^F = I$$

- The goal of the good type is to minimize costs due to informational asymmetries (underpricing)
- That is, good type promises to repay R_u^S in case of success and R_u^F in case of failure such that the break-even condition is satisfied and his/her utility is maximized

Pecking-order hypothesis

- The utility of the good type is

$$p(R^S - R_u^S) + (1-p)(R^F - R_u^F)$$

$$= \underbrace{pR^S + (1-p)R^F - I}_{NPV} + \underbrace{I}_{\substack{\text{amount raised} \\ \text{from investors}}} - \underbrace{[pR_u^S + (1-p)R_u^F]}_{\substack{\text{actual value of} \\ \text{the investors' claim}}}$$

$$= \underbrace{pR^S + (1-p)R^F - I}_{NPV} + \underbrace{[mR_u^S + (1-m)R_u^F]}_{=I} - \underbrace{[pR_u^S + (1-p)R_u^F]}_{\substack{\text{actual value of} \\ \text{the investors' claim}}}$$

$$= \underbrace{pR^S + (1-p)R^F - I}_{NPV} + \underbrace{[R_u^F + m(R_u^S - R_u^F)]}_{=I} - \underbrace{[R_u^F + p(R_u^S - R_u^F)]}_{\substack{\text{actual value of} \\ \text{the investors' claim}}}$$

$$= \underbrace{pR^S + (1-p)R^F - I}_{NPV} + \underbrace{(m-p)(R_u^S - R_u^F)}_{\substack{\text{discount due to} \\ \text{asymmetric information}}}$$

Pecking-order hypothesis

- The quantity $R_u^S - R_u^F$ is minimized under a debt contract with face value $F \geq R^F$
- That is, $R_u^F = R^F$ and R_u^S is such that

$$mR_u^S + (1-m)R^F = I \iff R_u^S = R^F + \frac{I - R^F}{m}$$

- This is a debt contract with face value R_u^S
- Alternatively, the entrepreneur issues riskless debt with face value R^F and complements the remainder with an equity issuance
- As bad types do not want to separate themselves from good types, both types end up preferring debt to equity when financing externally

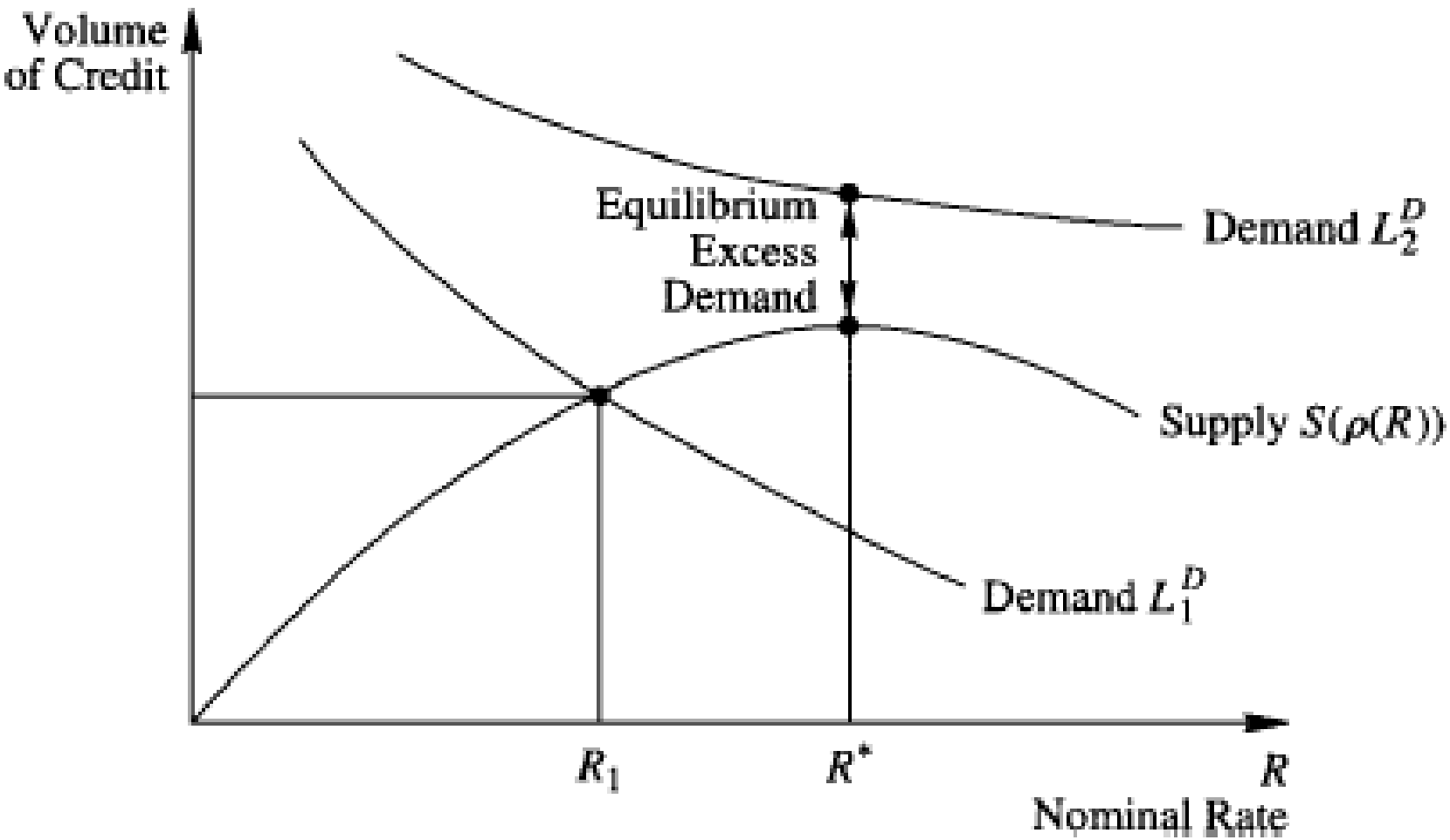
Stiglitz and Weiss (1981)

- Seminal work that studies rationing under imperfect information
- Notion of risk: mean-preserving spreads
  - Projects have the same mean
  - A mean-preserving spread of a random variable X can be obtained by adding a zero-mean random variable to each of the possible realizations of X
- Asymmetric information and adverse selection may cause the supply curve to be backward-bending for high interest rates

Stiglitz and Weiss (1981)

- The reason is that the pool of borrowers gets worse once interest rates increase, depleting lenders' returns and affecting supply
  - Think of a situation in which bad-quality borrowers have a higher probability of default
  - As the payoff in the default state does not depend on the level of interest rates, a marginal increase in interest rates affects bad borrowers less than good borrowers
- As a result, supply and demand curves might not intersect
- In this case, in equilibrium demand might exceed supply at the prevailing interest rate
- In other words, there is credit rationing

Stiglitz and Weiss (1981)



Stiglitz and Weiss (1981): example with two types

- Low risk: project yields R_L with probability p_L and zero otherwise
- High risk: project yields R_H with probability p_H and zero otherwise
- Assumptions
  - The probability of success is higher for the low-risk type: $p_L > p_H$
  - Alternatively, the probability of failure (default) is higher for the high-risk type: $1 - p_H > 1 - p_L$
  - The projects have the same expected return: $p_L R_L = p_H R_H = \bar{R}$ (same mean)
- We say that the high-risk project is a mean-preserving spread of the low-risk project

Stiglitz and Weiss (1981): example with two types

- Borrowers and lenders are risk-neutral; types are private information
- The project requires an investment of I
- If lenders require an interest rate of r, the borrowers' expected payoffs are
  - Low risk: $p_L[R_L - (1+\hat{r})I]) = \bar{R} - p_L(1+r)I$
  - High risk: $p_H[R_H - (1+\hat{r})I] = \bar{R} - p_H(1+r)I$
- As $p_L > p_H$, note that the expected payoff is smaller for the low-risk type: they have to repay $(1+r)I$ with a higher probability
- An increase in interest rates affects less high-risk borrowers

Stiglitz and Weiss (1981): example with two types

- There exists a $r^{\max}$ for which neither type applies when $r > r^{\max}$
- $r^{\max}$ is defined by

$$\bar{R} - p_H(1 + r^{\max})I = 0 \quad \Longleftrightarrow \quad r^{\max} = \frac{\bar{R}}{p_H I} - 1$$

- There exists a r^* such that when $r^* < r < r^{\max}$ only high risk borrowers apply for a loan
- r^* is defined by

$$\bar{R} - p_L(1 + r^*)I = 0 \quad \Longleftrightarrow \quad r^* = \frac{\bar{R}}{p_L I} - 1$$

- When $r < r^*$, both types apply for a loan

Stiglitz and Weiss (1981): example with two types

- When $r^* < r < r^{max}$, the lender's expected payoff is

$$\bar{p}_H(1+r)$$

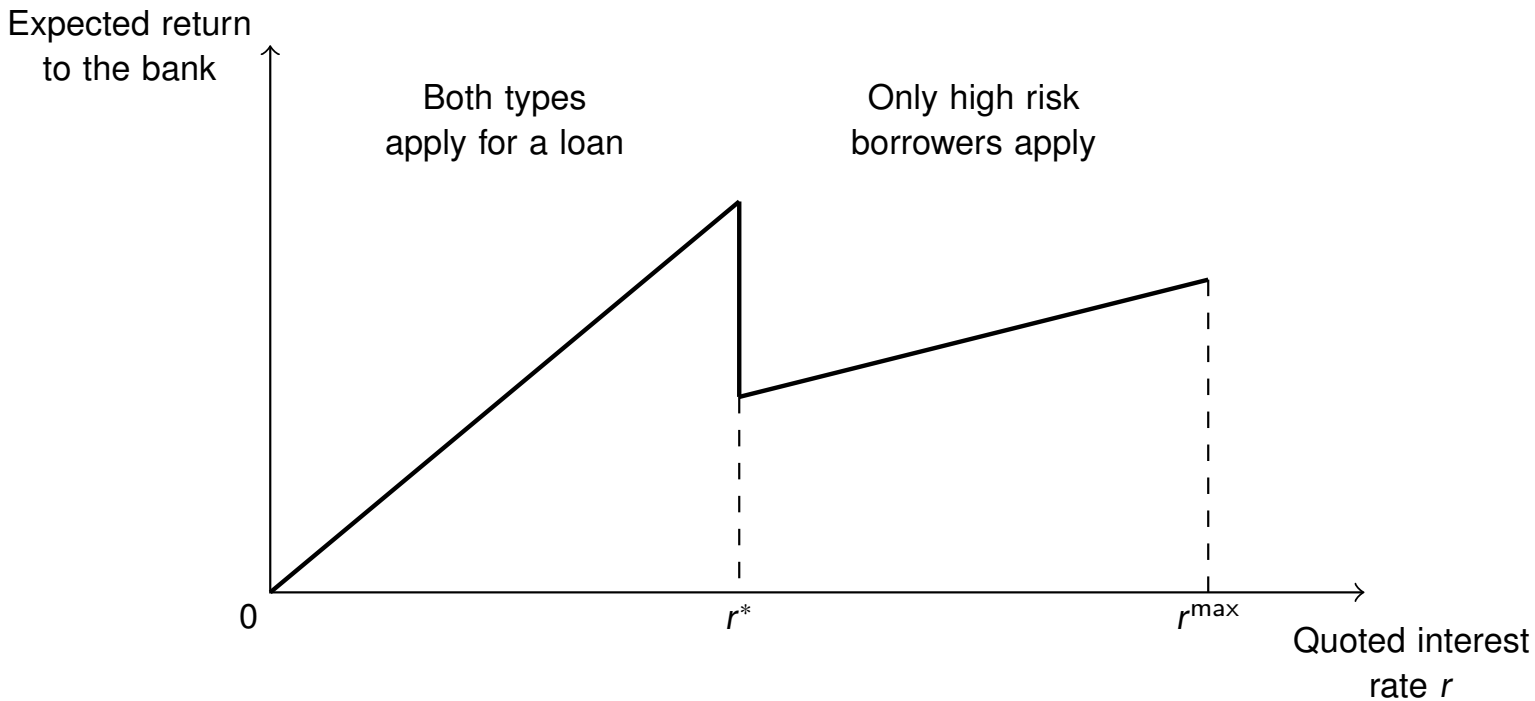
- If α is the proportion of low-risk types in the population, When $r < r^*$, the lender's expected payoff is

$$m(1+r) \quad \text{where} \quad m = \alpha p_L + (1-\alpha)p_H$$

- Remark: the expected return drops sharply around r^*, as the good types leave the market

- Expected return is not a monotonic function of interest rates

Stiglitz and Weiss (1981): example with two types



Stiglitz and Weiss (1981): numerical example

- $p_L = 0.9$, $X_L = 100$, $p_H = 0.6$, $X_H = 150$, $\alpha = 0.5$, $I = 85$
- We would have that $r^{\max} = 76.5\%$ and $r^* = 17.6\%$

Signaling

- In a market with adverse selection, low-risk borrowers might want to signal their types in an attempt to distinguish themselves from bad types
- Can good borrowers take **observable** actions that signal their repayment ability, that is, convince the uninformed side of the market about their types?
- To be a successful “separating” mechanism, the cost of the signaling device has to differ sufficiently among the types

Example: costly collateral pledging

- As before, we assume that
  - Projects require an investment of I and generates R in case of success and 0 otherwise
  - The good type entrepreneur has a probability of success of p, the bad type entrepreneur has a probability of success of q
  - The share of good types is α, and the prior about the probability of success of a rando borrower is $m = \alpha p + (1 - \alpha)q$
- Both types generate a positive NPV: $pR > I$, $qR > I$
  - Under asymmetric information, there is lending but the good type would like to minimize cross-subsidization

Example: costly collateral pledging

- Borrowers still have no cash on hand but they can pledge an (illiquid) asset as collateral
- Assets pledged as collateral are worth more when in the hands of the borrowers
- Equivalently, we can assume the existence of liquidation costs
- Assets that are worth C in the hands of borrowers are worth only δC in the lenders' hands, where $0 < \delta < 1$
- The use of collateral is inefficient: there is a loss of $(1 - \delta)C$ in case of default

Costly collateral pledging under symmetric information

- Under symmetric information there is no collateral pledging. Why?
- Consider the case of the good borrower. Suppose he/she pledges an amount $C > 0$ as collateral
- Competition $\Rightarrow$ lenders make zero profits $\Rightarrow$ the repayment R_u in case of success satisfies

$$pR_u + (1-p)\delta C = I \quad \Rightarrow \quad R_u = \frac{I}{p} - \frac{(1-p)\delta C}{p}$$

Costly collateral pledging under symmetric information

- The gain in utility of the good type is then

$$p(R - R_u) + (1-p)(-C) = \underbrace{pR - I}_{NPV} - \underbrace{(1-p)(1-\delta)C}_{\substack{\text{with probability } 1-p \\ (1-\delta)C \text{ is lost} \\ \text{(inefficiency)}}}$$

- The best choice is to choose $C = 0$

- Same argument for the bad type

Costly collateral pledging under asymmetric information

- Recall that under asymmetric information without the possibility of collateral pledging:

$$\underbrace{R_u}_{\substack{\text{Repayment of} \\ \text{both types under} \\ \text{asymmetric information}}} = \frac{I}{m} > \frac{I}{p} = \underbrace{R_u^G}_{\substack{\text{Repayment of the} \\ \text{good type under} \\ \text{symmetric information}}}$$

- Can the good type distinguish himself from the bad type by pledging collateral?

- We refer to a situation in which it is optimal for different types to choose different contracts as a **separating equilibrium**

- This outcome might be possible as the marginal cost of pledging collateral is higher for the bad type

Costly collateral pledging under asymmetric information

- In a separating equilibrium, each type chooses a different contract
  - Good type: R_u^G, C^G
  - Bad type: R_u^B, C^B
- Differently from a **pooling equilibrium**, in which different types choose the same contract, in a separating equilibrium lenders can infer the types with certainty by observing contract choices:
  - When they observe that a borrower chooses the contract of the bad type, R_u^B, C^B, they know that the probability of success is q
  - When they observe that a borrower chooses the contract of the good type, R_u^G, C^G, they know that the probability of success is p

Costly collateral pledging

- In a separating equilibrium, it is not in the interest of the bad type to pretend to be a good type:
  - The contract R_u^B, C^B yields a higher utility than the contract R_u^G, C^G
- If mimicking the contract choice of the good type is not in the interest of the bad type, the optimal choice of the bad type is not to pledge collateral
  - $C^B = 0$
  - Lenders will infer that the individual that does not post collateral has a probability of success of q and then the zero-profit repayment of the bad type is $R_u^B = I/q$, which implies $R_u^B = I/q$

Costly collateral pledging

- Suppose that the good type chooses a contract $\{\tilde{R}, \tilde{C}\}$, where $\tilde{R}$ is the repayment to the lender in case of success and $\tilde{C}$ is the collateral pledged

- The no-mimicking condition is

$$\underbrace{q(R - \tilde{R}) + (1 - q)(-\tilde{C})}_{\substack{\text{utility of the bad type} \\ \text{when he/she pretends} \\ \text{to be a good type}}} \leq \underbrace{q(R - R_u^B)}_{\substack{\text{utility of the bad type} \\ \text{when he/she does not pretend} \\ \text{to be a good type}}} = qR - I$$

Costly collateral pledging

- What is the optimal collateral choice of the good type that is consistent with lenders making zero profits and bad types not having incentives to lie?

- This is the solution of

$$\max_{\tilde{R},\tilde{C}} \; p(R-\tilde{R}) + (1-p)(-\tilde{C})$$

subject to

$$q(R-\tilde{R}) + (1-q)(-\tilde{C}) \leq qR - I \quad \textbf{(no-mimicking condition)}$$

$$p\tilde{R} + (1-p)(\delta\tilde{C}) \geq I \quad \textbf{(lender's break-even condition)}$$

Costly collateral pledging

- The solution to this problem is (both constraints are binding)

$$R_u^G = \frac{(1-q) - \delta(1-p)}{p(1-q) - \delta q(1-p)} I$$

and

$$C^G = \frac{I}{1 + \frac{q(1-p)(1-\delta)}{p-q}}$$

Costly collateral pledging

- If the adverse selection cost is very large (e.g. large proportion of bad borrowers, $1 - \alpha$, or large distance between p and q) the good type is better off with the possibility of signaling

$$\underbrace{p(R - R_u^G) + (1-p)(-C^G)}_{\substack{\text{utility under asymmetric} \\ \text{information } \textbf{with} \text{ the} \\ \text{possibility of signaling}}} \geq \underbrace{p(R - \frac{I}{m})}_{\substack{\text{utility under asymmetric} \\ \text{information } \textbf{without} \text{ the} \\ \text{possibility of signaling}}}$$

- In this case, signaling can reduce the "burden" of asymmetric information on the good type

Other signaling devices

- Payout policy
- Short-term maturities
- Underdiversification (self-financing)

Empirics: example

- Kawai et al. (2022): *We study how signaling affects equilibrium outcomes and welfare in an online credit market, using detailed data on loan characteristics and borrower repayment. We build and estimate an equilibrium model in which a borrower may signal her default risk through the reserve interest rate. Comparing markets with and without signaling relative to the benchmark with no asymmetric information, we find that adverse selection destroys as much as 34% of total surplus, up to 78% of which can be restored with signaling. We also estimate backward-bending supply curves for some markets, consistent with the prediction made by Stiglitz and Weiss in 1981.*

Empirics: example

- Michaely et al. (2021): *Contrary to signaling models' central predictions, changes in the level of cash flows do not empirically follow changes in dividends. We use the Campbell (1991) decomposition to construct cash-flow and discount-rate news from returns and find the following: (1) both dividend changes and repurchase announcements signal changes in cash-flow volatility (in opposite directions); (2) larger cash-flow volatility changes come with larger announcement returns; and (3) neither discount-rate news, nor the level of cash-flow news, nor total stock return volatility change following dividend changes. We conclude cash-flow news—and not discount-rate news—drive payout policy, and payout policy conveys information about future cash-flow volatility.*

Signaling through self-financing (optional)

- Entrepreneurs might seek external financing even though they have enough wealth to invest in their projects
- The reason is risk-sharing: risk-averse entrepreneurs may want to diversify their portfolio and sell part of their shares in the firm
- Large number of entrepreneurs, each endowed with a risky project
- All projects require an investment of 1
- Investors are risk-neutral and capital markets are competitive
- The project of a type-θ entrepreneur yields a random (gross) return of $\tilde{R}(\theta) = 1 + \tilde{r}(\theta)$, where $\tilde{r}(\theta) \sim N(\theta, \sigma^2)$

- The type θ is privately known by the entrepreneurs but its distribution in the population is common knowledge

- Entrepreneurs have enough initial wealth to finance their projects ($W_0 > 1$)

- Entrepreneurs are risk-averse with utility $u(w) = -e^{-\rho w}$, where w is their final wealth and $\rho = -u''/u' > 0$ is their (constant) coefficient of absolute risk-aversion

- There is no discounting and the outside option of investors has a zero net return

- Note: recall that if $X \sim N(\mu, \sigma^2)$, the random variable $Y = e^X$ has a log-normal distribution with mean

$$\mathbb{E}[Y] = e^{\mu + \frac{\sigma^2}{2}}$$

Symmetric information case

- Suppose that θ is observable and the type-θ entrepreneur sells a share $1 - \alpha$ of the project (or self-finances a share α)

- Competitive capital markets imply that outside investors just break even, that is, they obtain a share $1 - \alpha$ of the project by investing (paying)

$$(1-\alpha) \times \underbrace{\mathbb{E}[1+\tilde{r}(\theta)]}_{\text{expected value of the project}} = (1-\alpha)(1+\theta)$$

- After raising this amount, entrepreneurs invest $1 - (1-\alpha)(1+\theta)$ and their wealth is given

$$\begin{aligned} w &= W_0 - \underbrace{[1-(1-\alpha)(1+\theta)]}_{\text{entrepreneur's investment}} + \underbrace{\alpha}_{\text{entrepreneur's share}} \times \underbrace{[1+\tilde{r}(\theta)]}_{\text{gross return}} \\ &= W_0 + (1-\alpha)\theta + \alpha\tilde{r}(\theta) \end{aligned}$$

- If $\alpha = 1$, the entrepreneur bears all the costs (full self-financing),

$$w = W_0 + \tilde{r}(\theta) = W_0 - \underbrace{1}_{\text{investment}} + \underbrace{1 + \tilde{r}(\theta)}_{\text{gross return}}$$

- If $\alpha = 0$, the entrepreneur just sells the entire project (no self-financing). In this case, the entrepreneur "invests" $-\theta$ (sells the project for θ), that is, investors pay all the investment cost 1 and pay the entrepreneur θ for the project / idea

$$w = W_0 + \underbrace{\theta}_{\text{price of the project}}$$

- Note that when $\alpha = 0$, the entrepreneur is perfectly insured (constant wealth across states).

- For a general α, the entrepreneur's expected utility is (using the properties of the log-normal distribution)

$$\begin{aligned}
\mathbb{E}[u(w)] &= \mathbb{E}\Big[u\Big(W_0 + (1-\alpha)\theta + \alpha\tilde{r}(\theta)\Big)\Big] \\
&= \mathbb{E}\Big[-\rho e^{-\rho[W_0+(1-\alpha)\theta+\alpha\tilde{r}(\theta)]}\Big] \\
&= -\rho e^{-\rho[W_0+(1-\alpha)\theta+\alpha\theta-\frac{\rho\alpha^2\sigma^2}{2}]} \\
&= u\Big(W_0 + \theta - \underbrace{\frac{\rho\alpha^2\sigma^2}{2}}_{\substack{\text{disutility cost} \\ \text{of risk-bearing}}}\Big)
\end{aligned}$$

- Since the expected utility is decreasing in α, the risk-averse entrepreneur sells the entire project ($\alpha = 0$) and is perfectly insured

- Risk-neutral investors bear all the risk

θ is unobservable

- Suppose that entrepreneurs can choose either to self-finance ($\alpha = 1$) or to sell the entire project ($\alpha = 0$) at a price P common to all types
- When a type-θ entrepreneur self-finances, they obtain utility

$$u\left(W_0 + \theta - \frac{\rho\sigma^2}{2}\right)$$

- When they sell at a price P, they obtain utility $u(W_0 + P)$
- At a price P, a type-θ entrepreneurs seeks external financing if and only if

$$W_0 + P > W_0 + \theta - \frac{\rho\sigma^2}{2} \iff \theta < \hat{\theta} = P + \frac{\rho\sigma^2}{2}$$

- Adverse selection: only low-quality entrepreneurs with $\theta < \hat{\theta}$ seek financing
- Equilibrium price: $P = \mathbb{E}[\theta|\theta < \hat{\theta}]$

- This equilibrium can be inefficient
- Example: two types $1 < \theta_1 < \theta_2$, and $P(\theta = \theta_2) = \lambda$
- In the first-best, both types should sell the entire project
- If both types sell their project when θ is unobservable, the price would satisfy $P = \lambda\theta_2 + (1 - \lambda)\theta_1$
- The high-ability entrepreneur would be willing to sell the entire project if and only if

$$\lambda\theta_2 + (1 - \lambda)\theta_1 \geq \theta_2 - \frac{\rho\sigma^2}{2}$$

or

$$\underbrace{\theta_2 - [\lambda\theta_2 + (1 - \lambda)\theta_1]}_{\text{price discount}} \leq \underbrace{\rho\sigma^2/2}_{\substack{\text{disutility cost} \\ \text{of risk-bearing}}}$$

- If this condition is not satisfied the equilibrium is inefficient

Partial self-finance as a signal (Leland and Pyle, 1977)

- Suppose now that the entrepreneur can choose any $\alpha \in [0, 1]$
- Can (partial) self-financing be a signal to separate good entrepreneurs (θ_2) from bad ones (θ_1)?

Definition

A PBE is a set of self-financing and pricing strategies $\{\alpha(\theta), P(\alpha)\}$ *and a belief function* $\mu(\alpha)$ *giving the investors' common belief that the entrepreneur has high quality after observing* α *such that*

(i) Entrepreneur chooses α *optimally given the investors' strategies*

(ii) $\mu(\alpha)$ *is derived from Bayes' Rule where possible*

(iii) Investors' pricing strategies constitute a Nash equilibrium of the simultaneous-move game in which the probability $Pr(\theta = \theta_2|\alpha) = \mu(\alpha)$

- Note: change in the meaning of *P*. Before, I used *P* to denote the price of the project before the investment of 1 is made. Now *P* stands for the value of the "firm" after 1 is invested. Therefore, when the entrepreneur sells the project, he receives *P* – 1

- For a given pair (α, P), the expected utility of the entrepreneur is

$$
\begin{aligned}
v(\alpha, P|\theta) &= \mathbb{E}[u(w)] \\
&= \mathbb{E}\Big[u\Big(W_0 - [1 - (1-\alpha)P] + \alpha[1 + \tilde{r}(\theta)]\Big)\Big] \\
&= u\Big(W_0 - 1 + (1-\alpha)P + \alpha(1+\theta) - \frac{\rho\alpha^2\sigma^2}{2}\Big) \\
&= u\Big(W_0 - 1 + P - \alpha\underbrace{[P - (1+\theta)]}_{\text{mispricing}} - \frac{\rho\alpha^2\sigma^2}{2}\Big) \\
&= u\Big(W_0 - 1 + P \underbrace{-\alpha[P - (1+\theta)] - \frac{\rho\alpha^2\sigma^2}{2}}_{\text{cost of self-financing}}\Big)
\end{aligned}
$$

- Note that the marginal cost of self-financing is smaller for the high type

Constructing a separating equilibrium

- $P(\alpha) = \mu(\alpha)(1+\theta_2) + (1-\mu(\alpha))(1+\theta_1) \in [1+\theta_1, 1+\theta_2]$
- In a separating equilibrium, $P^*(\alpha^*(\theta_1)) = 1+\theta_1$ and $P^*(\alpha^*(\theta_2)) = 1+\theta_2$
- In a separating equilibrium, $\alpha^*(\theta_1) = 0$
- Denote by $\tilde{\alpha}$ the share of self-financing that makes bad entrepreneurs indifferent between their allocation and the allocation of good ones:

$$W_0 + \theta_1 = W_0 + \theta_2[1-\tilde{\alpha}] + \tilde{\alpha}\theta_1 - \frac{\rho\tilde{\alpha}^2\sigma^2}{2}$$

or

$$\frac{\tilde{\alpha}^2}{1-\tilde{\alpha}} = \frac{2(\theta_2-\theta_1)}{\rho\sigma^2}$$

- Note that $\alpha^*(\theta_2) \geq \tilde{\alpha}$, because otherwise bad entrepreneurs would prefer the allocation of good ones

- And good entrepreneurs prefer their allocation over that allocation of the bad entrepreneur if $\alpha^*(\theta_2) \leq \alpha'$, where

$$W_0 + \theta_2 - \frac{\rho\alpha'\sigma^2}{2} \geq W_0 + \theta_1$$

or

$$\alpha'^2 = \frac{2(\theta_2 - \theta_1)}{\rho\sigma^2}$$

- $\alpha' > \tilde{\alpha}$
- Again, there is a continuum of equilibria that can be Pareto ranked
- The Pareto-dominating equilibrium is characterized by $\alpha^*(\theta_2) = \tilde{\alpha}$

- A possible set of beliefs is given by $\mu(\alpha) = 1$ when $\alpha = \alpha^*(\theta_2)$ and $\mu(\alpha) = 0$ otherwise

- In the Pareto-dominating equilibrium, the utility of the bad entrepreneur is the same as when types are observable

$$u(W_0 + \theta_1)$$

and the utility of the good borrower is

$$u\Big(W_0 + \theta_2 - \underbrace{\frac{\rho\tilde{\alpha}^2\sigma^2}{2}}_{\text{informational cost}}\Big)$$

IPO underpricing

- Empirical evidence: IPOs tend to be underpriced
- Alternatively, prices tend to rise on the offer day
- Many potential explanations (see Lowry et al. (2017) for a survey)
- Here: asymmetric information among investors

IPO underpricing

- Firm that is issuing stocks has value $V \in \{V_B, V_G\}$, where $V_B < V_G$
- The share of good firms in the economy is α and is public knowledge
- The current owner of the firm does not know the true value of the firm
- Two types of buyers:
  - **Uninformed investors** believe the firm value is $\hat{V} = \alpha V_G + (1 - \alpha) V_B$ to it
  - **Informed investors (specialists)** know the true value of the firm

IPO underpricing

- Informed investors have capital λA and uninformed investors have capital $(1 - \lambda)A$
- Assumption: informed investors do not have enough capital to fund the IPO by themselves:

$$(1 - \lambda)A > V_G > V_B > \lambda A$$

- In case the offer is oversubscribed, investors are rationed on a pro-rata basis

IPO underpricing

- When informed and uninformed investors are free to trade, and the order flow is public information, the expected market-clearing equilibrium price has to be $\hat{V}$
- Why?
  - When $V = V_G$, informed investors will want to hold the stock if $P < V_G$ and everyone infers the true price is V_G
  - When $V = V_B$, informed investors will not hold the stock if $P > V_B$ and everyone infers the price is V_B
- As the firm is good with probability α, the expected market price after the issuance will be $\mathbb{E}[P_1] = \hat{V}$

IPO underpricing

- Let $P_0 \in (V_B, V_G)$ be the offer price?
- Is $P_0 = \mathbb{E}[P_1] = \hat{V}$?
  - If $P_0 < \mathbb{E}[P_1]$, the IPO will be underpriced (on average)
- Informed investors' demand
  - If $V = V_G$: they know the offer is underpriced and demand $D_I^G = \lambda A / P_0$
  - If $V = V_B$: they know the offer is overpriced and demand $D_I^B = 0$

IPO underpricing

- Uninformed investors' demand
  - Is not contingent on the true value of $V \Rightarrow$ either $D_U = 0$ or $D_U = \frac{(1-\lambda)A}{P_0}$
  - If $D_U = 0$, the IPO fails (insufficient demand)
  - When $V = V_G$ and $D_u = \frac{(1-\lambda)A}{P_0}$, uninformed investors only get a fraction $\frac{D_U}{D_U + D_I^G} = (1 - \lambda)$ of the firm
  - When $V = V_B$ and $D_u = \frac{(1-\lambda)A}{P_0}$, they buy the whole issue (**winners' curse**)

IPO underpricing

- When is $D_U = \frac{(1-\lambda)A}{P_0}$?
- Uninformed investors need to break even on average

$$\alpha \underbrace{(1-\lambda)}_{\substack{\text{share owned if} \\ \text{the firm is good}}} \underbrace{(V_G - P_0)}_{\substack{\text{profit if the} \\ \text{firm is good}}} + (1-\alpha) \underbrace{1}_{\substack{\text{share owned if} \\ \text{the firm is bad}}} \underbrace{(V_B - P_0)}_{\substack{\text{loss if the} \\ \text{firm is bad}}} = 0$$

$$\iff \hat{V} - P_0 - \alpha\lambda(V_G - P_0) = 0$$

$$\iff P_0 = \frac{\hat{V} - \lambda\alpha V_G}{1-\lambda\alpha} = \hat{V}\frac{1 - \lambda\alpha V_G/\hat{V}}{1-\lambda\alpha} < \hat{V} = \mathbb{E}[P_1] \quad (\text{as } \hat{V} < V_G)$$

- Despite the underpricing, uninformed investors only break even on average

IPO underpricing

- Underpricing disappears under symmetric information ($\lambda = 0$)
- Underpricing increases with the fraction of informed investors: $\frac{\partial P_0}{\partial \lambda} < 0$
- Underpricing can increase if the underwriter favors large, informed buyers
- Keeping $\hat{V}$ constant, underpricing is increases when $V_G - V_B$ increases

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